

# Flagged LLT Polynomials

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ICERM ECR Seminar

joint work with Jonah Blasiak, Mark Haiman, Jennifer Morse, and Anna Pun

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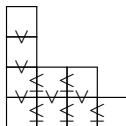
# Young Tableaux

## Definition

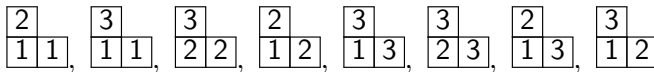
Filling of partition diagram of  $\lambda$  with numbers such that

- 1 strictly increasing up columns
- 2 weakly increasing along rows

Collection is called  $\text{SSYT}(\lambda)$ .



For  $\lambda = (2, 1)$ ,



# Schur polynomials

Associate a polynomial to  $\text{SSYT}(\lambda)$ .

$$\begin{array}{cccccccc} \begin{array}{|c|c|} \hline 2 & \\ \hline 1 & 1 \\ \hline \end{array}, & \begin{array}{|c|c|} \hline 3 & \\ \hline 1 & 1 \\ \hline \end{array}, & \begin{array}{|c|c|} \hline 3 & \\ \hline 2 & 2 \\ \hline \end{array}, & \begin{array}{|c|c|} \hline 2 & \\ \hline 1 & 2 \\ \hline \end{array}, & \begin{array}{|c|c|} \hline 3 & \\ \hline 1 & 3 \\ \hline \end{array}, & \begin{array}{|c|c|} \hline 3 & \\ \hline 2 & 3 \\ \hline \end{array}, & \begin{array}{|c|c|} \hline 2 & \\ \hline 1 & 3 \\ \hline \end{array}, & \begin{array}{|c|c|} \hline 3 & \\ \hline 1 & 2 \\ \hline \end{array} \\ \rightarrow & \begin{array}{|c|c|} \hline z_2 & \\ \hline z_1 & z_1 \\ \hline \end{array}, & \begin{array}{|c|c|} \hline z_3 & \\ \hline z_1 & z_1 \\ \hline \end{array}, & \begin{array}{|c|c|} \hline z_3 & \\ \hline z_2 & z_2 \\ \hline \end{array}, & \begin{array}{|c|c|} \hline z_2 & \\ \hline z_1 & z_2 \\ \hline \end{array}, & \begin{array}{|c|c|} \hline z_3 & \\ \hline z_1 & z_3 \\ \hline \end{array}, & \begin{array}{|c|c|} \hline z_3 & \\ \hline z_2 & z_3 \\ \hline \end{array}, & \begin{array}{|c|c|} \hline z_2 & \\ \hline z_1 & z_3 \\ \hline \end{array}, & \begin{array}{|c|c|} \hline z_3 & \\ \hline z_1 & z_2 \\ \hline \end{array} \end{array}$$

$$s_{(2,1)}(z_1, z_2, z_3) = z_1^2 z_2 + z_1^2 z_3 + z_2^2 z_3 + z_1 z_2^2 + z_1 z_3^2 + z_2 z_3^2 + 2z_1 z_2 z_3$$

## Definition

For  $\lambda$  a partition

$$s_\lambda = \sum_{T \in \text{SSYT}(\lambda)} \mathbf{z}^T \text{ for } \mathbf{z}^T = \prod_{i \in T} z_i$$

- $s_\lambda$  is a symmetric function.
- $\{s_\lambda\}_\lambda$  forms a basis for  $\Lambda$ .

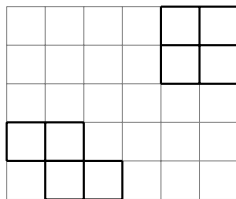
# Products of Schur polynomials

- Littlewood-Richardson rule:  $s_\lambda s_\mu = \sum_\nu c_{\lambda\mu}^\nu s_\nu$  for  $c_{\lambda\mu}^\nu \in \mathbb{N}$ .
- Skew Schur function:  $s_{\nu/\lambda} = \sum_{T \in \text{SSYT}(\nu/\lambda)} x^T = \sum_\mu c_{\lambda\mu}^\nu s_\mu$ .
- Straightforward observation:  $s_{\nu/\lambda} s_{\kappa/\mu} = \sum_{\substack{T \in \text{SSYT}(\nu/\lambda) \\ S \in \text{SSYT}(\kappa/\mu)}} x^T x^S$ .
- $q$ -deformation?

# Key Object: LLT Polynomials

Let  $\nu = (\nu_{(1)}, \dots, \nu_{(k)})$  be a tuple of skew shapes. (Skew shape =  $\lambda \setminus \mu$ )

$$\nu = \left( \begin{array}{|c|c|c|} \hline \square & \square & \\ \hline \square & \square & \square \\ \hline \end{array}, \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \end{array} \right)$$



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|    |    |    |    |   |   |
|----|----|----|----|---|---|
| -4 | -3 | -2 | -1 | 0 | 1 |
| -3 | -2 | -1 | 0  | 1 | 2 |
| -2 | -1 | 0  | 1  | 2 | 3 |
| -1 | 0  | 1  | 2  | 3 | 4 |
| 0  | 1  | 2  | 3  | 4 | 5 |

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Attacking pairs:  $(b_2, b_3), (b_3, b_4), (b_4, b_5), (b_4, b_6), (b_5, b_7), (b_6, b_7), (b_7, b_8)$



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- A *semistandard tableau* on  $\nu$  is a map  $T: \nu \rightarrow \mathbb{Z}_+$  which restricts to a semistandard tableau on each  $\nu_{(i)}$ .

The *LLT polynomial* indexed by a tuple of skew shapes  $\nu$  is

$$\mathcal{G}_\nu(\mathbf{z}; q) = \sum_{T \in \text{SSYT}(\nu)} \mathbf{z}^T,$$

$$\mathbf{z}^T = \prod_{a \in \nu} z_{T(a)}.$$

$T =$

|   |   |   |  |   |   |
|---|---|---|--|---|---|
|   |   |   |  | 5 | 6 |
|   |   |   |  | 1 | 1 |
|   |   |   |  |   |   |
| 2 | 4 |   |  |   |   |
|   | 3 | 5 |  |   |   |

$$\mathbf{z}^T = z_1^2 z_2 z_3 z_4 z_5^2 z_6$$



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where  $\text{inv}(T)$  is the number of attacking inversions in  $T$  and  $\mathbf{z}^T = \prod_{a \in \nu} z_{T(a)}$ .

$T =$

|   |   |   |  |   |   |
|---|---|---|--|---|---|
|   |   |   |  | 5 | 6 |
|   |   |   |  | 1 | 1 |
|   |   |   |  |   |   |
| 2 | 4 |   |  |   |   |
|   | 3 | 5 |  |   |   |

non-inversion

$$\mathbf{z}^T = z_1^2 z_2 z_3 z_4 z_5^2 z_6$$

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$$\text{inv}(T) = 4, \quad \mathbf{z}^T = z_1^2 z_2 z_3 z_4 z_5^2 z_6$$

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- When  $\nu^{(i)}$  are partitions, the Schur-expansion coefficients are essentially parabolic Kazhdan-Luzstig polynomials.
- $\mathcal{G}_\nu$  is Schur-positive for any tuple of skew shapes  $\nu$  [Grojnowski-Haiman, 2007].

# Some notable occurrences of LLT Polynomials

- Haglund-Haiman-Loehr formula for Macdonald polynomials:

$$\tilde{H}_{\mu}(X; q, t) = t^{n(\mu)} \sum_R \left( \prod_{\substack{\square \\ \overline{u}}} q^{a+1} t^l \right) \mathcal{G}_R(X; t^{-1}).$$

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- For  $G$  = incomparability graph for natural unit interval order (encoded by Dyck path  $P$ ),

$$\chi_G(x; t) = (1 - t)^{-|V|} \mathcal{G}_{\nu(P)}[(1 - t)x; t].$$

# Flagged Tableaux

- Fix partition shape  $\lambda = (\lambda_1, \dots, \lambda_l)$  and *flag*  
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- Denote the set of such tableaux via  $\text{FT}(\lambda, \mathbf{b})$ .
- E.g.,  $\lambda = (2, 1)$ ,  $\mathbf{b} = (1, 3)$ ,  $\text{FT}(\lambda, \mathbf{b}) =$

$$\begin{array}{|c|c|} \hline 2 & \leq 3 \\ \hline 1 & 1 \\ \hline \end{array} \leq 1 \quad \begin{array}{|c|c|} \hline 3 & \leq 3 \\ \hline 1 & 1 \\ \hline \end{array} \leq 1$$

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- Note, there is no issue with letting length of  $\mathbf{b}$  be less than  $l$ .

# Flagged Schur functions (Lascoux-Schützenberger, Wachs)

- For partition  $\lambda$  and flag  $\mathbf{b}$ , the *flagged Schur function* is given by

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- $\mathfrak{S}_w(x) = s_{\lambda, \mathbf{b}}(x)$  for *vexillary*  $w \in S_n$  (2143-avoiding).
- $s_{\lambda, \mathbf{b}}(x)$  are examples of *Demazure characters*.

## Demazure characters and atoms

The *Demazure operator*  $\pi_i$  acts on  $f \in \mathbb{Q}(q, t)[x_1^{\pm 1}, \dots, x_N^{\pm 1}]$  by

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The *Demazure characters* or *key polynomials* are constructed from

- $\mathcal{D}_\lambda = x^\lambda := x_1^{\lambda_1} \cdots x_N^{\lambda_N}$  for partition  $\lambda$ .
- $\mathcal{D}_{s_i(\alpha)} = \pi_i \mathcal{D}_\alpha$  for  $\alpha_i > \alpha_{i+1}$ , for any  $\alpha \in \mathbb{N}^N$ .

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*Demazure atoms* are defined the same as keys but with  $\hat{\pi}_i := \pi_i - 1$  in place of  $\pi_i$ :

- $\mathcal{A}_\lambda = x^\lambda$  for partition  $\lambda$ .
- $\mathcal{A}_{s_i(\alpha)} = \hat{\pi}_i \mathcal{A}_\alpha$  for  $\alpha_i > \alpha_{i+1}$ , for any  $\alpha \in \mathbb{N}^N$ .

# Examples

$$\mathcal{D}_{520} = x_1^5 x_2^2$$

$$\mathcal{D}_{250} = \pi_1 \mathcal{D}_{520} = \pi_1(x_1^5 x_2^2) = x_1^5 x_2^2 + x_1^4 x_2^3 + x_1^3 x_2^4 + x_1^2 x_2^5$$

$$\mathcal{D}_{205} = \pi_2 \mathcal{D}_{250} = \pi_2(x_1^5 x_2^2 + x_1^4 x_2^3 + x_1^3 x_2^4 + x_1^2 x_2^5)$$

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$$\mathcal{D}_{520} = \mathcal{A}_{520}$$

$$\mathcal{D}_{250} = \mathcal{A}_{520} + \mathcal{A}_{250}$$

$$\mathcal{D}_{205} = \mathcal{A}_{520} + \mathcal{A}_{250} + \mathcal{A}_{502} + \mathcal{A}_{205}$$

$$\vdots$$

# Recovering Symmetric Functions

- If  $\mathbf{b} = (n, \dots, n)$  for  $n = |\lambda|$ , then  $s_{\lambda, \mathbf{b}}(x) = s_{\lambda}(x_1, \dots, x_n)$ .

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- For  $w = s_{i_1} s_{i_2} \cdots s_{i_m} \in S_N$  reduced,  $\pi_w := \pi_{i_1} \pi_{i_2} \cdots \pi_{i_m}$ .

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- For *Weyl symmetrization operator*  $\pi_{w_0}$ , we have

$$\pi_{w_0}(s_{\lambda, \mathbf{b}}(x)) = s_{\lambda}(x), \quad \pi_{w_0}(\mathcal{D}_{\alpha}) = \mathcal{D}_{\text{sort}(\alpha)} = s_{\alpha+},$$

$$\pi_{w_0}(\mathcal{A}_{\alpha}) = \begin{cases} s_{\alpha} & \alpha \text{ a partition,} \\ 0 & \text{else.} \end{cases}$$

# Flagged LLT Polynomials (Blasiak-Haiman-Morse-Pun-S.)

- Let  $e_1, \dots, e_l$  be the row ends of  $\nu$ , ordered in reverse reading order.
- Fix flag  $\mathbf{b} = (b_1 \leq \dots \leq b_l)$ .
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|  |       |       |  |  |  |       |
|--|-------|-------|--|--|--|-------|
|  |       |       |  |  |  | $e_3$ |
|  |       |       |  |  |  | $e_1$ |
|  |       |       |  |  |  |       |
|  | $e_4$ |       |  |  |  |       |
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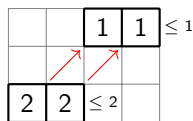
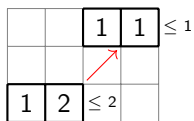
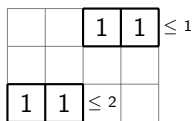
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|   |   |   |  |   |   |          |
|---|---|---|--|---|---|----------|
|   |   |   |  | 2 | 3 | $\leq 3$ |
|   |   |   |  | 1 | 1 | $\leq 1$ |
|   |   |   |  |   |   |          |
| 2 | 4 |   |  |   |   | $\leq 4$ |
|   | 1 | 2 |  |   |   | $\leq 2$ |

# Flagged LLT Polynomials

$$\nu = (\square\square, \square\square), \mathbf{b} = (1, 2)$$

T



$$\mathcal{G}_{r,\nu}(x; t) = x_1^4 + t x_1^3 x_2 + t^2 x_1^2 x_2^2$$

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- $\mathcal{G}_{\nu, \mathbf{b}}(x; t)$  have an algebraic formula via “nonsymmetric Hall-Littlewood polynomials.”
- $\mathcal{G}_{\nu, \mathbf{b}}(x; t)$  are conjecturally positive in terms of *Demazure atoms*.

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  - unbarred letters weakly increase in rows, strictly increase in columns.
  - barred letters strictly increase in rows, weakly increase in columns.
  - $T(e_i) \leq b_i$  for  $i = 1, \dots, l$ .

# Signed flagged LLT polynomials

- Signed alphabet  $\mathcal{A} = 1 < \bar{1} < 2 < \bar{2} \dots$
- $\text{FT}^\pm(\nu, \mathbf{b})$  = fillings of  $\nu$  from  $\mathcal{A}$  satisfying
  - unbarred letters weakly increase in rows, strictly increase in columns.
  - barred letters strictly increase in rows, weakly increase in columns.
  - $T(e_i) \leq b_i$  for  $i = 1, \dots, l$ .

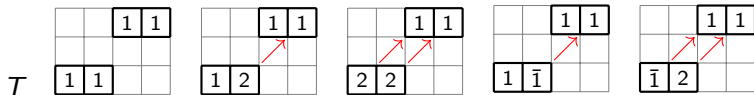
The *signed flagged LLT polynomial* indexed by  $\nu$  and flag  $\mathbf{b}$  is

$$\mathcal{G}_{\nu, \mathbf{b}}^\pm(\mathbf{x}; t) = \sum_{T \in \text{FT}^\pm(\nu, \mathbf{b})} t^{\text{inv}(T)} (-t)^{-\#\text{bar}(T)} \mathbf{x}^{|T|},$$

where  $|T|$  is the result of removing all bars from  $T$ .

# Signed flagged LLT polynomials

$$\nu = (\square\square, \square\square), \mathbf{b} = (1, 2)$$



$$\begin{aligned} \mathcal{G}_{\nu, \mathbf{b}}^{\pm}(\mathbf{x}; t) &= x_1^4 + tx_1^3x_2 + t^2x_1^2x_2^2 - x_1^4 - tx_1^3x_2 \\ &= t^2x_1^2x_2^2 \end{aligned}$$

# Flagged plethysm

Define *flagged plethysm*  $\Pi_{t,x}: \mathbb{K}[x_1, \dots, x_l] \rightarrow \mathbb{K}[x_1, \dots, x_l]$  to be the (over-determined) linear map

$$\Pi_{t,x} \mathcal{G}_{\nu, \mathbf{b}}^{\pm}(x; t^{-1}) = \mathcal{G}_{\nu, \mathbf{b}}(x; t^{-1}).$$

**Theorem (Blasiak-Haiman-Morse-Pun-S., 2025+)**

$\Pi_{t,x}$  is well-defined.

Note,  $\Pi_{t,x}$  is a “nonsymmetric analogue” of the plethystic map  $f[X] \mapsto f[X/(1-t)]$  for symmetric function  $f$ .

# Applications

- ① Can define *modified  $r$ -nonsymmetric Macdonald polynomials* (or *modern Macdonald polynomials*) via a *flagged* Haglund-Haiman-Loehr formula:

$$H_{\eta|\lambda}(x; q, t) = t^n \sum_R \left( \prod_{\substack{U \\ \square}} q^{a+1} t^l \right) \mathcal{G}_{R, (b_1, \dots, b_r)}(x; t^{-1}).$$

These symmetrize to Macdonald  $H$ -polynomials and are conjecturally Demazure atom positive.

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- 2 Using  $H_{\eta|\lambda}$  to define a nonsymmetric analogue of  $\nabla$ , we show a “nonsymmetric shuffle theorem” expressed in terms of flagged LLT polynomials associated to flagged Dyck paths.
- 3 Tewari-Wilson-Zhang define chromatic nonsymmetric polynomials associated to  $d \times d$  Dyck paths starting with  $r$  north steps,  $\chi_{\mathbf{b}, \pi}$ . We show

$$\chi_{(1, \dots, r), \pi}(x; t) = (1 - t)^{r-d} \mathcal{G}_{\nu(\pi), (1, \dots, r)}^{\pm}(x; t).$$